



Deliberation layer

LLM planner
(semantic, open-ended, sec-min cycles)

task analysis · planning · goal specification

goals
(G_k, w_k)

goal interface
(G_k, w_k); cost c

feedback
(obs, r)



Execution layer

small models: RL / rule
(continuous control, ms-s cycles)

reactive control · tool use · low-level skills

cross-domain adversarial scenario



Conditional Layering Law

$$\Delta V = -c + f(D_1', B_{if}) \times \text{headroom}(V_m)$$

Layering pays iff:

- (i) D_1' : core difficulty in LLM-strong decision forms
- (ii) $B_{if} > 0$: goal interface carries information the executor cannot reconstruct
- (iii) $\text{headroom}(V_m) > 0$: monolithic baseline is not saturated

Falsifiable predictions

P1 D_1' fails $\Rightarrow \Delta V < 0$, regardless of interface

P2 $\text{headroom} \approx 0 \Rightarrow$ layering can only lose

P3 positive region when all three hold; absent otherwise

TEST 1 · Boundary (grid: all three fail)

- strong-hold: +0.457 / +0.537 (CIs > 0)
- Deployed hybrids: $\Delta V_A = -0.213$; $\Delta V_B = -0.240$
- Three rescue channels fail (P1)

failure lessons $\Rightarrow$ gates

TESTS 2 & 3 · Positive region (14-scenario suite)

- Gates: D_1' / D2 / D3; D1 measured, not gated
- Layered 0.77-0.78 vs 0.63-0.72
- Beats Rule 11/14; RL 12/14
- End-to-end LLM 0.43; behind 14/14



Probe: LLM discriminator < rule ($C_{info} < 0$), yet layering wins $\Rightarrow$ semantic goal decomposition.

**ANSWER:** The LLM belongs at the planning layer, above a mature executor, only where decision-form match, interface carrying, and baseline headroom jointly hold — never end-to-end.